

COMPUTER GRAPHICS

Unit-I: Introduction to Computer Graphics

Comprehensive Study Notes

1. Definition of Computer Graphics and Its Applications

Definition

Computer Graphics is the field of computer science that deals with generating, manipulating, and displaying visual content using computers. It involves the creation, storage, and manipulation of images and models of objects using mathematical algorithms and data structures.

Classification of Applications

Category	Applications
Creation / Production	3D Modeling, Animation, Rendering, Game Development
Analysis / Visualization	Data Visualization, Medical Imaging, Scientific Visualization, Engineering
Presentation	UI Design, Advertising, Business, Education

- **3D Modeling** – Creating three-dimensional digital representations of objects.
- **Animation** – Generating moving images by sequencing frames.
- **Rendering** – Generating photorealistic images from 3D models.
- **Game Development** – Creating interactive visual experiences.
- **Data Visualization** – Representing data graphically for analysis.

- **Medical Imaging** – MRI, CT scans, ultrasound visualization.
- **Scientific Visualization** – Simulating scientific phenomena visually.
- **Engineering** – CAD, structural analysis, simulations.
- **UI Design** – Designing graphical user interfaces.
- **Advertising** – Commercial graphics and promotional content.
- **Business** – Charts, graphs, presentations.
- **Education** – Interactive learning materials, simulations.

2. Video Display Devices

2.1 Cathode Ray Tube (CRT) Displays

A CRT display uses an electron beam to illuminate phosphor-coated screen. The beam is controlled by magnetic deflection coils and scans across the screen to create images.

Raster Scan Display

- The electron beam sweeps across the screen from top to bottom, left to right, one row at a time (raster pattern).
- Screen is divided into a matrix of pixels arranged in rows (scan lines).
- Each pixel's intensity is stored in a **Frame Buffer** (refresh buffer).
- **Refresh Rate**: Typically 60–120 Hz (times per second the screen is redrawn).
- **Interlacing**: Alternates between odd and even scan lines to reduce flicker.
- Used in: Traditional TVs, computer monitors.

Random Scan Display (Vector Display)

- The electron beam is directed only to specific points where the picture is to be drawn.
- Draws lines and shapes directly (vector graphics).
- Also called **Calligraphic Display**.
- Produces smooth lines without jagged edges.
- Limited to line drawings; cannot display filled areas easily.
- Used in: Oscilloscopes, early CAD systems.

Feature	Raster Scan	Random Scan
Beam movement	Entire screen row by row	Only where picture exists
Image type	Pixel-based (bitmap)	Line-based (vector)
Resolution	Limited by pixel count	High (limited by beam)
Refresh	Full screen each cycle	Only drawn parts

Cost	Lower	Higher
Jaggies	Visible (aliasing)	Smooth lines
Use	TVs, monitors	Oscilloscopes, plotters

2.2 Color CRT Monitors

- Uses three electron guns: **Red, Green, Blue**.
- **Shadow Mask:** A metal plate with tiny holes that aligns electron beams to the correct phosphor dots.
- Each pixel consists of three phosphor dots (RGB).
- **Delta Electron Gun Arrangement:** Three guns arranged in a triangular pattern.
- **In-line Gun Arrangement:** Three guns arranged horizontally (modern CRTs).

2.3 Direct View Storage Tube (DVST)

- Uses a storage mesh to retain the image, eliminating the need for constant refreshing.
- The image is drawn once and stored until intentionally erased.
- **Advantages:** No flicker, high resolution, no refresh buffer needed.
- **Disadvantages:** No color, no selective erasing, slow update.
- Used in: Early graphics terminals.

2.4 Flat Panel Displays

Type	Technology	Characteristics
LCD	Liquid Crystal Display	Thin, low power, good quality, common in monitors & laptops. Uses liquid crystals that block or pass light when electric current is applied.
LED	Light Emitting Diode	Uses LEDs as backlight for LCD panels. Thinner, brighter, more energy efficient than CCFL-backlit LCD. Better contrast ratios.
Plasma	Plasma Display Panel	Uses small cells containing ionized gas (plasma) that emits UV light to excite phosphors. Good color, wide viewing angle, but higher power consumption.

OLED	Organic Light Emitting Diode	Each pixel emits its own light (no backlight). Excellent contrast (true blacks), very thin, low power, flexible displays possible.
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3. Input Devices

Device	Function	Use Cases
Keyboard	Text input, commands	General computing, programming
Mouse	Pointing, selection, navigation	GUI interaction, design
Joystick	Directional control	Gaming, simulation, robotics
Trackball	Precision pointing	CAD, kiosks, specialized environments
Digitizer	Converts analog drawings to digital coordinates	CAD, engineering drawings, map digitization
Scanner	Captures document/image as digital data	Document digitization, OCR, archiving
Touch Screen	Direct touch interaction	Smartphones, tablets, kiosks, ATMs
Light Pen	Screen selection using light detection	CAD (older systems), point-of-sale

Keyboard Types

- **Mechanical:** Individual mechanical switches. Durable, tactile feedback.
- **Membrane:** Pressure pads. Cheap, quiet, less durable.
- **Virtual/On-screen:** Software-based, touch screens.
- **Ergonomic:** Designed to reduce strain.
- **Gaming:** Programmable keys, RGB lighting, fast response.

Mouse Types

- **Mechanical:** Rolling ball detects motion.
- **Optical:** LED sensor detects surface movement. More accurate.
- **Laser:** Laser sensor for higher precision.
- **Wireless:** Bluetooth/RF connectivity.

Joystick Types

- **Digital:** Simple directional switches (on/off).
- **Analog:** Variable resistance for proportional control.
- **Force-feedback:** Provides haptic feedback.

Digitizer

- Converts analog position information into digital coordinates.
- Uses electromagnetic induction, resistive, or ultrasonic sensing.
- Applications: CAD, GIS, medical imaging.

Image Scanner

- **Flatbed:** Document placed on glass; scanning head moves.
- **Sheet-fed:** Document moves past stationary sensor.
- **Drum:** High resolution; document wrapped around drum.
- **Handheld:** Portable, manual scanning.
- **OCR:** Optical Character Recognition technology.

Touch Screen Technologies

- **Resistive:** Pressure-sensitive. Durable, low cost.
- **Capacitive:** Detects electrical properties of skin. Multi-touch support.
- **Infrared:** IR beams detect touch. High durability.
- **Surface Acoustic Wave:** Ultrasonic waves detect touch.

4. Output Devices

Printers

Type	Technology	Characteristics
Laser Printer	Electrophotographic; uses laser beam to draw image on drum, toner applied and fused to paper.	Fast, high quality, low cost per page for text. Common in offices.
Inkjet Printer	Sprays tiny droplets of ink onto paper through microscopic nozzles.	Good color quality, affordable for home use. Slower than laser. Better for photos.

Plotters

- Vector-based output device using pens to draw continuous lines.
- **Flatbed Plotter**: Paper stationary, pen moves in X-Y directions.
- **Drum Plotter**: Paper rolls on drum, pen moves horizontally.
- Used for: CAD drawings, architectural blueprints, engineering designs.
- Advantages: High precision, smooth lines, large format.

5. Graphics Software

General Purpose Graphics Software

- **Microsoft Paint**: Simple raster-based drawing tool. Basic image editing.
- **Adobe Photoshop**: Professional raster graphics editor. Photo manipulation, digital art, design. Supports layers, filters, masks.

Specialized Graphics Software

Software	Purpose	Key Features
AutoCAD	Computer-Aided Design (CAD)	2D/3D drafting, precision tools, architectural/engineering design. Industry standard for technical drawing.

Maya	3D Animation & Modeling	Professional 3D animation, modeling, simulation, rendering. Used in film, gaming, VFX.
Blender	Open-source 3D Creation	Modeling, sculpting, animation, rendering, video editing. Free and community-driven.

Quick Reference Summary – Unit I

Key Terms

1. **Computer Graphics** – Creation/manipulation of visual content using computers.
2. **Raster Scan** – Display scanning entire screen row by row.
3. **Random Scan** – Display drawing only picture points.
4. **Pixel** – Smallest addressable screen element.
5. **Frame Buffer** – Memory storing pixel data.
6. **Refresh Rate** – Times screen redrawn per second.
7. **Shadow Mask** – Metal plate aligning electron beams in CRT.
8. **DVST** – Direct View Storage Tube (stores image without refresh).
9. **LCD** – Liquid Crystal Display.
10. **LED** – Light Emitting Diode.
11. **OLED** – Organic Light Emitting Diode.
12. **Digitizer** – Converts analog to digital input.
13. **OCR** – Optical Character Recognition.

Display Comparison

Type	Characteristics
CRT	Bulky, high power, good color, older technology
LCD	Thin, low power, good quality, common
LED	Thin, medium power, bright, common
OLED	Very thin, low power, excellent quality, expensive
DVST	High resolution, no flicker, no color, obsolete

Input Device Functions

Device	Primary Function
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Keyboard	Text input, commands
Mouse	Pointing, selection, navigation
Trackball	Precision pointing, CAD
Joystick	Gaming, simulation, robotics
Digitizer	Drawing, CAD, 3D input
Scanner	Document/image capture
Touch Panel	Direct interaction, smartphones
Light Pen	Screen selection (CRT only)

Questions

- Q1.** Define Computer Graphics. Explain its applications in detail.
- Q2.** Differentiate between Raster Scan and Random Scan displays.
- Q3.** Explain the working of a Color CRT Monitor with a neat diagram.
- Q4.** What is DVST? Discuss its advantages and disadvantages.
- Q5.** Explain the various types of Flat Panel Displays (LCD, LED, Plasma, OLED).
- Q6.** Describe various input devices used in computer graphics.
- Q7.** Explain the working of a touch screen. Discuss different touch screen technologies.
- Q8.** Differentiate between Laser printer and Inkjet printer.
- Q9.** What are plotters? Explain flatbed and drum plotters.
- Q10.** Write a note on different types of graphics software (Paint, Photoshop, AutoCAD, Maya, Blender).
- Q11.** Explain the architecture of a raster scan system with a block diagram.
- Q12.** What is a frame buffer? Explain its role in raster graphics.

— End of Computer Graphics — Unit I Notes —